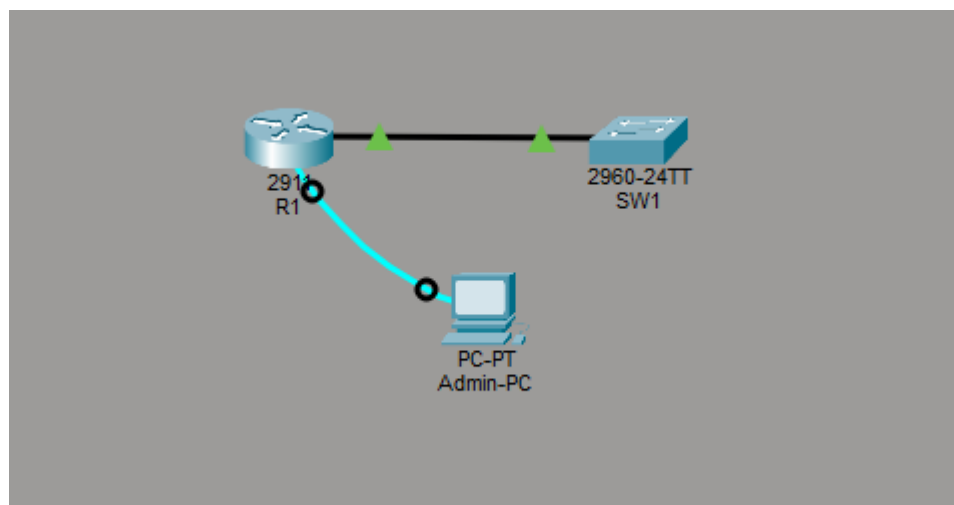


NET512 — Weekly Practical

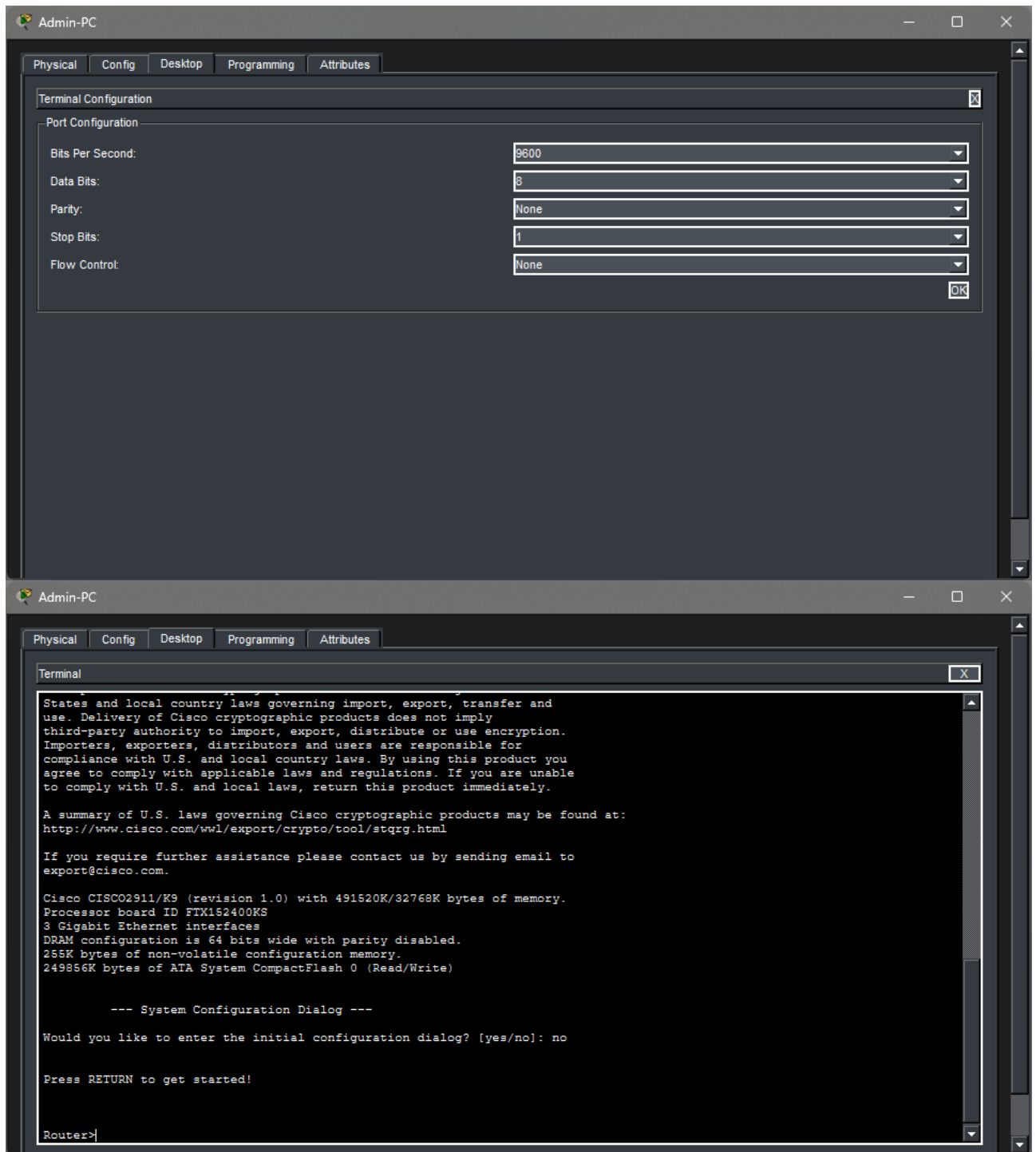
NAME	Christiaan van Wyk	WEEK	Week 1
STUDENT NO.	20261193	DATE	03 August 2026

Topology

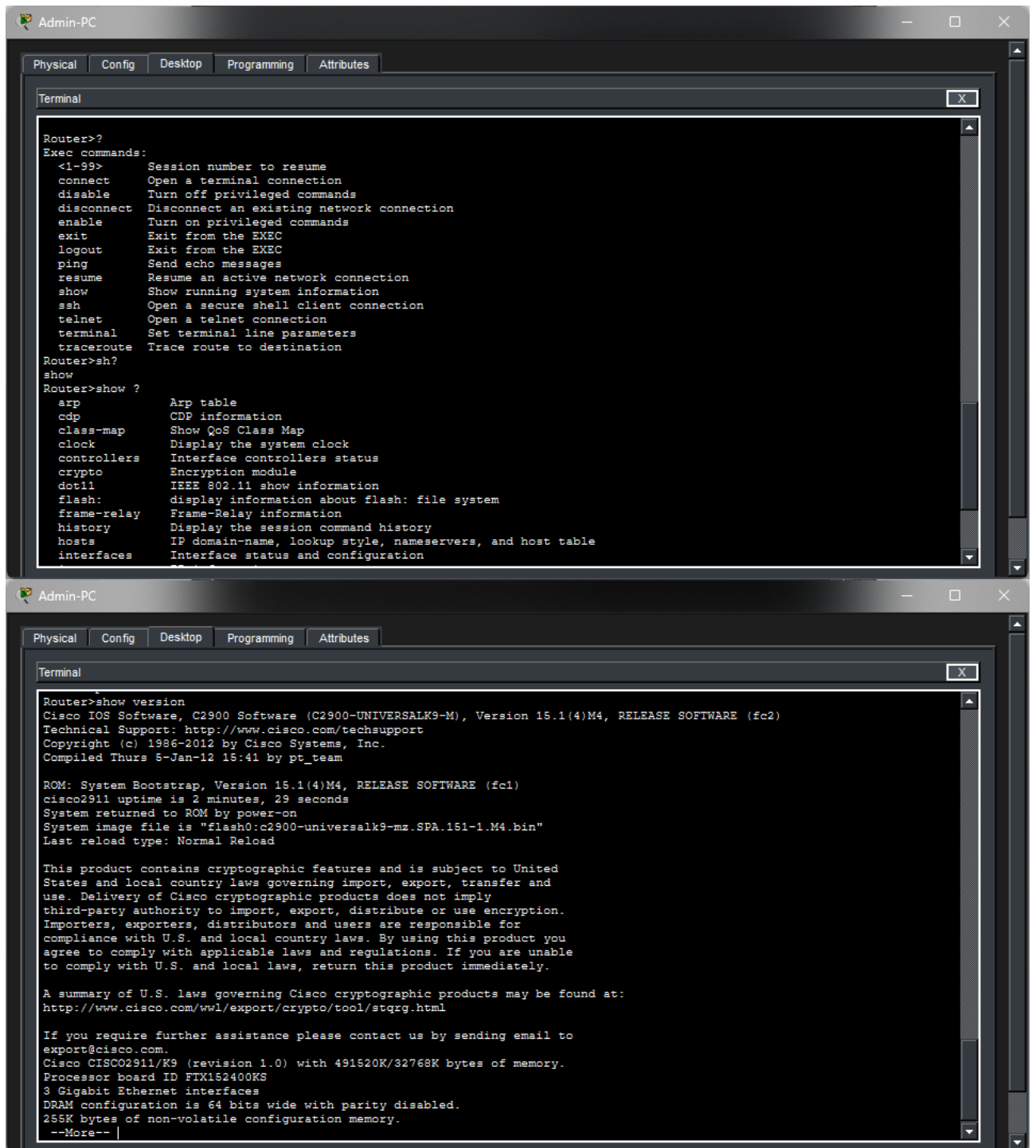


Tasks

Task 01 – Connect to device via console



Task 02 — Explore User EXEC mode



```
Admin-PC
Physical Config Desktop Programming Attributes
Terminal
Router>?
Exec commands:
<1-99>      Session number to resume
connect     Open a terminal connection
disable     Turn off privileged commands
disconnect  Disconnect an existing network connection
enable      Turn on privileged commands
exit        Exit from the EXEC
logout      Exit from the EXEC
ping        Send echo messages
resume      Resume an active network connection
show        Show running system information
ssh         Open a secure shell client connection
telnet      Open a telnet connection
terminal    Set terminal line parameters
traceroute  Trace route to destination
Router>sh?
show
Router>show ?
arp          Arp table
cdp          CDP information
class-map    Show QoS Class Map
clock        Display the system clock
controllers  Interface controllers status
crypto       Encryption module
dot11        IEEE 802.11 show information
flash:       display information about flash: file system
frame-relay  Frame-Relay information
history      Display the session command history
hosts        IP domain-name, lookup style, nameservers, and host table
interfaces   Interface status and configuration

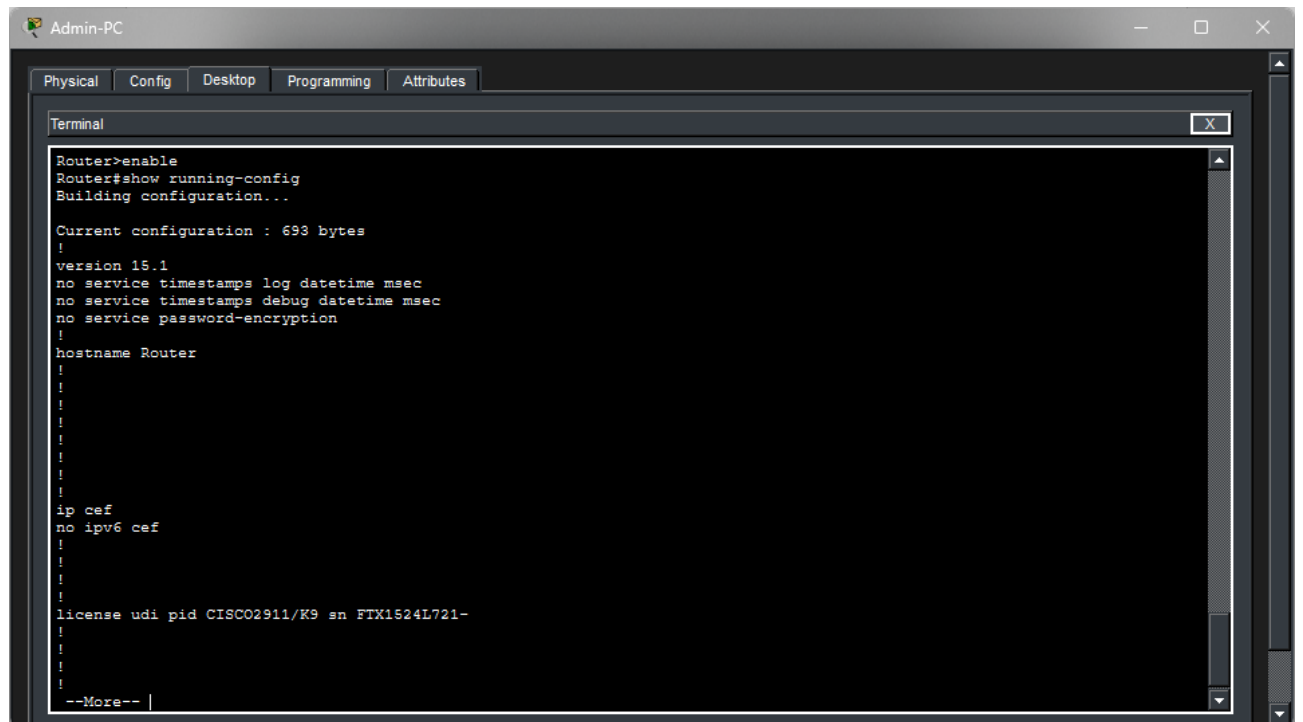
Admin-PC
Physical Config Desktop Programming Attributes
Terminal
Router>show version
Cisco IOS Software, C2900 Software (C2900-UNIVERSALK9-M), Version 15.1(4)M4, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2012 by Cisco Systems, Inc.
Compiled Thurs 5-Jan-12 15:41 by pt_team

ROM: System Bootstrap, Version 15.1(4)M4, RELEASE SOFTWARE (fc1)
cisco2911 uptime is 2 minutes, 29 seconds
System returned to ROM by power-on
System image file is "flash0:c2900-universalk9-mz.SPA.151-1.M4.bin"
Last reload type: Normal Reload

This product contains cryptographic features and is subject to United
States and local country laws governing import, export, transfer and
use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/ww1/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.
Cisco CISC02911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
256K bytes of non-volatile configuration memory.
--More--
```

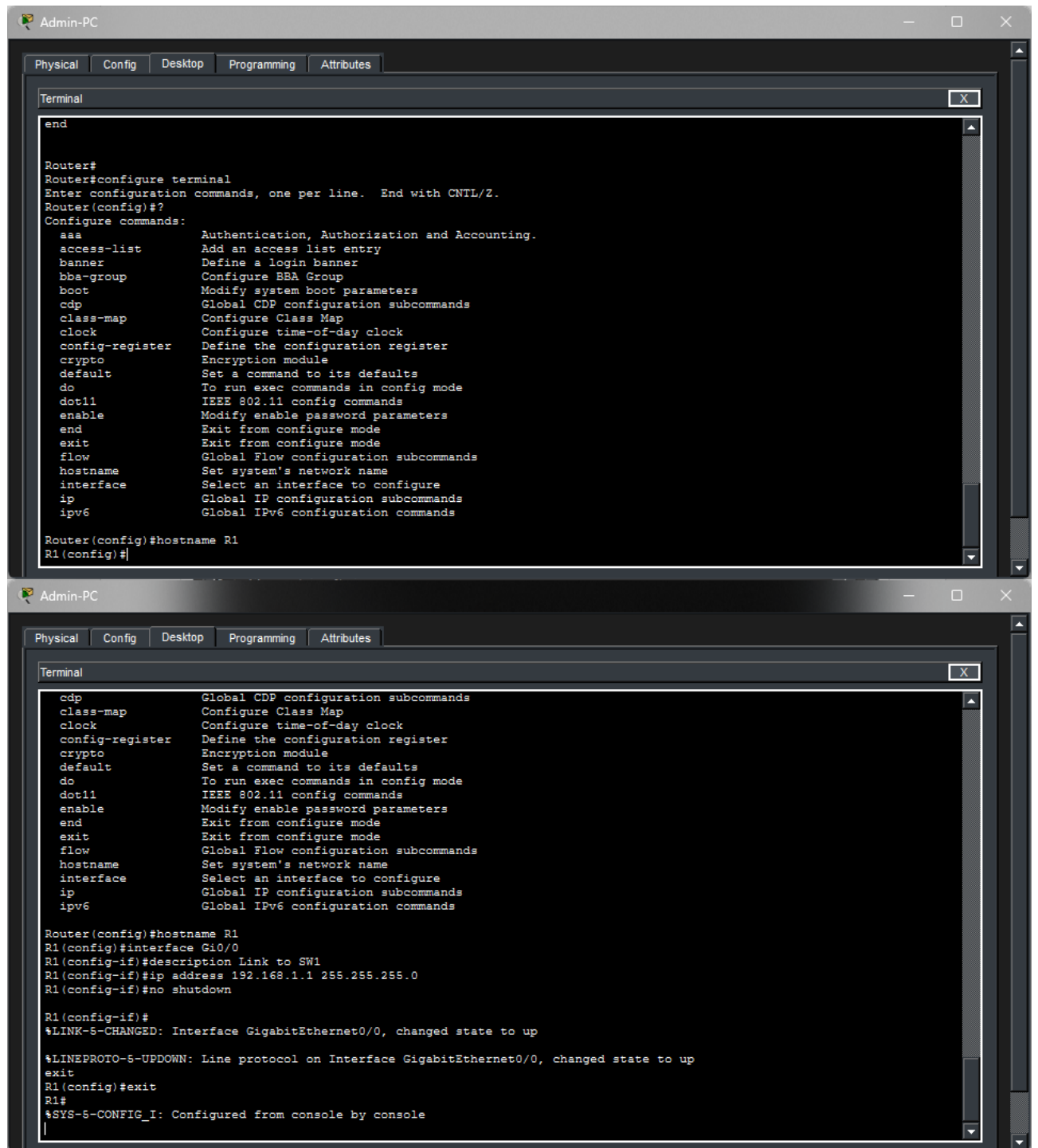


The image shows a terminal window titled "Admin-PC" with tabs for "Physical", "Config", "Desktop", "Programming", and "Attributes". The "Config" tab is active, and a terminal window is open within it. The terminal displays the following text:

```
Router>enable
Router#show running-config
Building configuration...

Current configuration : 693 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!
!
!
!
!
!
!
!
ip cef
no ipv6 cef
!
!
!
!
license udi pid CISCO2911/K9 sn FTX1524L721-
!
!
!
!
--More-- |
```

Task 03 — Global Configuration mode



The image shows two screenshots of a Cisco Admin-PC terminal window. The top screenshot shows the initial configuration steps, including entering the configuration terminal, listing available commands, and setting the hostname to R1. The bottom screenshot shows the configuration of interface Gi0/0, including setting the description, IP address, and enabling the interface.

```
Admin-PC
Physical Config Desktop Programming Attributes
Terminal
end

Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#?
Configure commands:
aaa          Authentication, Authorization and Accounting.
access-list  Add an access list entry
banner       Define a login banner
bba-group    Configure BBA Group
boot         Modify system boot parameters
cdp          Global CDP configuration subcommands
class-map    Configure Class Map
clock        Configure time-of-day clock
config-register Define the configuration register
crypto       Encryption module
default      Set a command to its defaults
do           To run exec commands in config mode
dot11        IEEE 802.11 config commands
enable       Modify enable password parameters
end          Exit from configure mode
exit         Exit from configure mode
flow         Global Flow configuration subcommands
hostname     Set system's network name
interface    Select an interface to configure
ip           Global IP configuration subcommands
ipv6         Global IPv6 configuration commands

Router(config)#hostname R1
R1(config)#

Admin-PC
Physical Config Desktop Programming Attributes
Terminal
cdp          Global CDP configuration subcommands
class-map    Configure Class Map
clock        Configure time-of-day clock
config-register Define the configuration register
crypto       Encryption module
default      Set a command to its defaults
do           To run exec commands in config mode
dot11        IEEE 802.11 config commands
enable       Modify enable password parameters
end          Exit from configure mode
exit         Exit from configure mode
flow         Global Flow configuration subcommands
hostname     Set system's network name
interface    Select an interface to configure
ip           Global IP configuration subcommands
ipv6         Global IPv6 configuration commands

Router(config)#hostname R1
R1(config)#interface Gi0/0
R1(config-if)#description Link to SW1
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console
```

Task 04 — IOS configuration management

The image displays two sequential screenshots of a Cisco IOS configuration session on a router named R1, accessed via a terminal window on an Admin-PC. The top screenshot shows the output of the `show running-config` command, displaying the current configuration, including version 15.1, service timestamps, hostname R1, and IP CEF. The bottom screenshot shows the sequence of commands to save the configuration: `show startup-config` (confirming it is not present), `copy running-config startup-config` (prompting for a destination filename), `write memory` (building configuration), and a final `show startup-config` (confirming the saved configuration matches the running configuration).

```
Admin-PC
Physical Config Desktop Programming Attributes
Terminal
R1#show running-config
Building configuration...

Current configuration : 727 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname R1
!
!
!
!
!
!
!
!
!
!
ip cef
no ipv6 cef
!
!
!
!
license udi pid CISCO2911/K9 sn FTX1524L721-
!
!
!
!
--More--

Admin-PC
Physical Config Desktop Programming Attributes
Terminal
R1#show startup-config
startup-config is not present
R1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R1#write memory
Building configuration...
[OK]
R1#show startup-config
Using 727 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname R1
!
!
!
!
!
!
!
!
!
!
ip cef
no ipv6 cef
!
!
!
!
```